Alexis Imbert

PhD Student

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Alexis-IMBERT
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Education

2021 – 2024	**Engineering Degree in Computer Science and Information Technology**, *INSA Rouen*
2023 – 2024	**Master's Degree in Data Science and Engineering**, *University of Rouen*, Rouen
2023	**Academic Exchange**, *Royal Melbourne Institute of Technology (RMIT)*, Melbourne, Australia Semester abroad, courses on AI for professionals, ethics in AI
2019 – 2021	**PEIP Program**, *Sorbonne University*, Paris General scientific preparatory program

Experience

2024 – 2027	**PhD Student**, *Sulcal Pit Analysis of Cortical Surfaces with GNN*, INSA Rouen Normandie
February – October 2024	**Research Engineer Internship**, *Historical Image Description with LLM*, Teklia, Paris

Skills

Machine Learning	Python, Scikit-learn (contributor), PyTorch, PyTorch-Geometric, Torch-Lightning, LLM, vLLM, YOLO - Ultralytics (contributor), Open-source contributions
Programming	Python, C, Java, Octave & Matlab, HTML, CSS, JavaScript, Node.js, Vue.js, AJAX, Flask, PHP, Django
Sports	Swimming, Climbing, Skiing, Snowboarding
Music	Musical training and classical guitar, solo and in groups for 13 years
Extracurricular	President of INSA Rouen and high school photography clubs, elected member of the ITI Department Council and INSA Rouen Board of Directors

Publications

Alexis IMBERT, Benoit GAUZERE, Sylvain TAKERKART, Guillaume AUZIAS, and Paul HONEINE. Insights on Using Graph Neural Networks forSulcal Graphs Predictive Models. June 2025.

Alexis IMBERT, Mélodie BOILLET, and Christopher KERMORVANT. Évaluation de Large Language Models pour la description de photographies, June 2024.

Certifications

English	B2 - Professional	*825/990 TOEIC*
French	Native	
Spanish	B2	

Research Focus

2024 – 2027 **PhD Student**, *Sulcal Pit Analysis of Cortical Surfaces with GNN*, INSA Rouen Normandie

Thesis supervised by Paul Honeine (University of Rouen), Guillaume Auzias (Institut de Neurosciences de la Timone, CNRS, Aix-Marseille University), and Benoît Gaüzère (INSA Rouen Normandie). Funded by the French National Research Agency as part of the FAMOUS project. **Research Questions:**

1. Hierarchical Information Analysis in Brain Graphs: How can graph machine learning models effectively capture and utilize hierarchical information in brain graphs to enhance the representation and analysis of cortical folding patterns?
2. Robustness to MRI Acquisition Variations: Can a graph machine learning model trained on brain data from one acquisition center generalize effectively to data from different centers, improving reproducibility in neuroimaging studies? This research explores transfer learning applications in GML.
3. Local Variation Identification for Cognitive Functions: How can GML methods identify and analyze local variations in brain anatomy, and what insights can these variations provide regarding cognitive functions and neurological conditions?

Publication at GBR 2025 [1].

February – October 2024 **Research Engineer Internship**, *Teklia*, Paris

Automatic description of historical Japanese photography using large multimodal language models (LLM). Project in collaboration with the Guimet Museum of Asian Arts and Teklia.

- Benchmarking of prompt engineering software
- Image processing for consistency
- Large-scale experiment deployment
- Fine-tuning of existing models with project images
- Poster presentation at SIFED 2024 [2]

Projects

2023 **INSA Certified Project**

Project in collaboration with Actémium Paris Transport. Object detection in images and videos within the compressed domain, applied to road surveillance videos in tunnels.

2023 **Web Platform for Academic Mobility Support**

Design of a web platform to facilitate administrative procedures for academic mobility from the perspectives of students and faculty.

2022 **Fermi and von Neumann Elephant**

Project illustrating Von Neumann's elephant concept, with functionalities for generating new designs, computing Fourier series, and reproducing initial designs with adjustable parameters.

2022 **"Pixel War" Multiplayer Game**

Recreation of the "Pixel War" multiplayer game in a team of three, including backend, frontend, and database development. Players collaboratively place pixels on a canvas to create drawings.

2021 **Recognition of Visually Identical Images**

Application of mathematical methods to detect pairs of visually identical images, accounting for variations in size, noise, and transformations. Introduction to machine learning concepts.

2021 **Spellchecker in C**

Team project in C to develop a spellchecker using tree structures.